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Plasticity of Gene Expression in Spaceflight and Postflight in Relation to Cardiovascular Disease: Mechanisms and Candidate Repurposed Drugs

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ABSTRACT

Spaceflight poses unique challenges to human health due to exposure to increased levels of cosmic radiation, microgravity, and associated oxidative stress. These environmental factors can lead to cellular damage, inflammation, and a range of health complications, including cardiovascular problems, immune system impairment, and an increased risk of cancer. Nuclear factor erythroid 2-related factor 2 (NRF2) is a critical transcription factor that regulates the body's defense mechanisms against oxidative stress by promoting the expression of antioxidant enzymes. Recent research has shed more light on the critical role of NRF2 in addressing space-related health challenges. In this study, we developed a computational methodology to explore the plasticity of the gene expression profile in flight and postflight conditions, highlighting the genes and corresponding mechanisms that do not return to ground levels and correlate with gene signatures associated with cardiovascular disease (CVD). RNA sequencing (RNA-seq) data from human induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs) have been used to investigate the cellular effects of microgravity on cardiac function. Gene expression monotonicity studies were performed and linked to genome-wide association studies (GWAS) to highlight the monotonically expressed genes associated with CVD. The selected monotonically expressed genes were also mapped onto the NRF2 network to investigate the impact of spaceflight on human cardiomyocyte function in the context of redox signaling pathways. Based on this knowledge, we used computational drug repurposing methods to suggest a short list of repurposed drug candidates that can be further tested in astronauts for the prevention of CVD. This study provides insights into the molecular and redox signaling alterations in cardiomyocytes induced by spaceflight, laying the foundation for future research aimed at mitigating cardiovascular risks in astronauts and advancing clinical applications on Earth.

1 | Introduction

Since the historic Apollo 11 moon landing in 1969, space travel has progressed remarkably, with numerous aerospace companies now pushing toward ambitious goals such as human habitation on Mars. As these commercial space programs advance, understanding the health impacts of space environments remains a significant challenge. Space travel presents unique risks, including exposure to microgravity, prolonged confinement, isolation, space radiation, and increased bacterial virulence [1]. Although current data are sparse, evidence suggests that astronauts may face distinct health issues compared to those living on Earth.

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Spaceflight induces a variety of physiological responses, including cardiovascular deficits, loss of bone and muscle mass, compromised immune function, chromosomal aberrations, and metabolic alterations [2, 3].

As future missions extend the duration of space exposure, the cardiovascular risks to astronauts are likely to increase, making it crucial to gain a comprehensive understanding of the molecular mechanisms underlying these physiological changes [4]. The NASA Twin Study revealed that extended exposure to microgravity lowers mean arterial pressure and elevates cardiac output [5]. Nonetheless, there is limited understanding of how microgravity affects human cardiac function at the cellular level.

One of the key challenges in spaceflight is managing oxidative stress, which disrupts critical oxidative pathways necessary for cellular homeostasis [6, 7]. Nuclear factor erythroid 2-related factor 2 (NRF2), a master regulator of oxidative defense, and its interactors play a pivotal role in responding to this stress. During spaceflight, significant alterations in the expression of genes involved in oxidative stress pathways have been observed, especially those related to cell cycle arrest and apoptosis [8]. A key player in these processes is CDKN1A/P21, which is upregulated in response to space-induced stress and works closely with NRF2 to coordinate cellular defense mechanisms. This interaction is crucial for managing oxidative damage across various tissues, emphasizing the central role of NRF2 and its associated pathways in countering the harmful effects of the space environment [4].

In this study, we developed a computational methodology aimed to explore the plasticity of the gene expression profile in flight and postflight conditions. To accomplish this, we utilized RNA sequencing (RNA-seq) data from human induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs) to identify monotonically expressed genes, that is, differentially expressed genes (DEGs) with ascending or descending trend in their differential expressions (log 2-fold changes [$\log_2$FC]) across flight and postflight condition. The list of monotonically expressed genes in the postflight level was then utilized in a genome-wide association studies (GWAS)-based analysis in order to identify those that are associated with cardiovascular disease (CVD). The monotonically expressed genes in postflight were also mapped to the NRF2 network (interactome and regulome) to investigate genes that are also associated with redox signaling pathways. At that point, an in silico drug repurposing pipeline highlighted candidate repurposed drugs that were experimentally shown to reverse the expression of the majority of the CVD-associated GWAS and redox-related monotonically expressed genes.

Our computational approach identified 343 monotonically expressed genes in postflight condition that (a) were further correlated with CVDs through GWAS-based analysis, (b) they were associated with oxidative stress and their interactions with NRF2. Our analysis ended up with 30 selected monotonically expressed genes and 10 candidate repurposed drugs for the postflight condition. To the best of our knowledge, no prior computational studies have specifically focused on the monotonically expressed genes in flight and postflight conditions to investigate the effects of spaceflight on human cardiac function.

2 | Methods

2.1 | Methodology Overview

The workflow of the proposed methodology is described in the following diagram (Figure 1):

- *Data selection and analysis*: Transcriptomic RNA-seq data from hiPSC-CMs was collected from the GeneLab database [9]. The preprocessed and already DEGs in flight and postflight conditions versus ground control were downloaded from the GeneLab Processed RNA-seq Files.
- *Identification of significant altered genes in flight and postflight condition*: We selected the significant DEGs for flight and postflight using as selection criterion p value < 0.05.
- *Monotonically expressed genes*: From these gene signatures, we investigated the common genes between the two conditions in order to identify the monotonically expressed genes (a monotonic trend either consistently increasing or decreasing $|\log_2 FC|$).
- *NRF2-related monotonically expressed genes in postflight*: We also found the monotonically expressed genes with higher $|\log_2 FC|$ in postflight that also belong to the NRF2 neighborhood.
- *CVD-related GWAS analysis on monotonically expressed genes in postflight*: Finally, we identified the monotonically expressed genes with higher $|\log_2 FCs|$ in postflight that are associated with a higher risk for CVDs.
- *Shortlist of candidate repurposed drugs*: The NRF2 and the CVD-related monotonically expressed genes in postflight were used as input in an in silico drug repurposing tool called Drug Gene Budger (DGB) [10]. We found repurposed drugs that have been experimentally confirmed to reverse the expression of these genes. Using a majority voting approach, we selected those drugs that reverse the expression of at least 60% of the genes.

2.2 | Collection and of Transcriptome Profiles

The transcriptome dataset of hiPSC-CMs used in this study was sourced from NASA's GeneLab platform (genelab.nasa.gov) [9]. Specifically, we accessed the dataset with OSD ID: OSD-258, titled "Effects of Spaceflight on Human Induced Pluripotent [11] Stem Cell-Derived Cardiomyocyte Structure and Function" [6]. The hiPSC-CMs were collected after 4.5 weeks in space, with an additional sample collected on the 10th day after their return to Earth. Ground-based hiPSC-CMs, collected at the same time point after return, served as controls.

We downloaded the differential expression analysis results from the GeneLab database and the GeneLab Processed RNA-Seq Files. We specifically examined the comparisons between (i) Space Flight & 4.5 weeks microgravity exposure versus Ground Control & 5.5 weeks plus 10 days, and (ii) Space Flight & 5.5 weeks microgravity exposure followed by 10 days of normal gravity versus Ground Control & 5.5 weeks plus 10 days. For each

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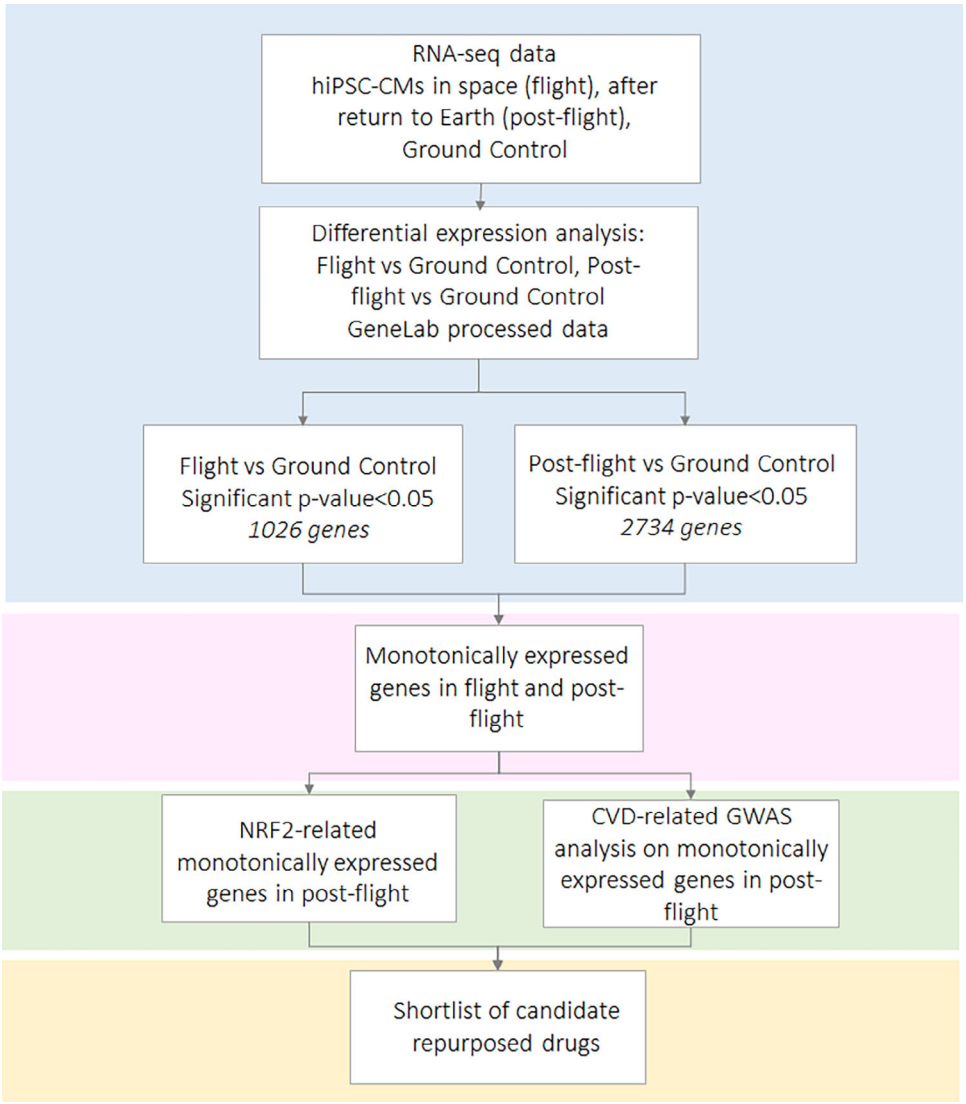


FIGURE 1 | Overview of the workflow.

comparison, we retrieved the $\log_2$FC and p values, selecting genes with p values < 0.05 as significant for each space condition.

2.3 | Selection of the NRF2 Neighborhood

The NRF2 network was constructed using data from the repository at (http://sbi.imim.es/data/nrf2/) [12]. This network integrates protein–protein interactions (PPIs) related to the NRF2-regulating pathway, using information from multiple curated resources such as IntAct, MINT, BioGRID, HPRD, KEGG, and PhosphoSite. These datasets were compiled into a comprehensive human interactome, forming the basis for mapping NRF2 interactions. Several well-characterized interactors, including KEAP1, β-TrCP, and MAF proteins, are incorporated into the network. The network also captures interactions with nuclear proteins that regulate gene expression, such as those linked to bZIP transcription factors, nuclear receptors, coactivators, and histone acetylation processes. These interactions reveal additional regulatory mechanisms beyond NRF2's direct activities. NRF2's phosphorylation at various residues enables interactions with kinases, such as GSK-3 and protein kinase C isoforms, which are regulated by upstream membrane receptors and scaffold proteins. Beyond physical interactions, the network highlights biologically enriched processes within NRF2's neighborhood, including pathways for the biosynthesis of pentose, tetrapyrrole, heme, glucose 6-phosphate, cysteine, glutathione (GSH), glyceraldehyde-3-phosphate, and NADPH. Many of these regulatory proteins are indirectly connected through intermediary proteins, creating a robust NRF2 neighborhood. This network integrates both physical and regulatory (transcriptional) interactions, linking *NFE2L2* (the NRF2 gene) to the products of its target genes.

2.4 | Genetic Investigation of the Monotonically Expressed Genes in Postflight Condition

Using the monotonically expressed genes in postflight condition, we explored the genetic associations in which the genes were involved in. Specifically, we utilized the GWAS catalog, a database that includes a collection of human GWAS [13], to explore the

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cardiovascular (CVD) phenotypes and related risk factors linked to the genes of interest and the variants within the genes affecting the risk for disease. We mapped the monotonically expressed genes to the reported gene of the GWAS catalog associations. We did not filter for any specific populations as our studied population can have a diverse ethnic background. A p value $< 5 \times 10–^{8}$ was used to identify the associations of genome-wide significance. We do not report the associations where the effect size was absent from the data. The analyses were performed in R software, version 4.3.3.

2.5 | Multisource Enrichment Analysis

We conducted a multisource enrichment analysis using the web tool Enrichr-KG (https://maayanlab.cloud/enrichr-kg) [14–16]. Enrichr-KG is a knowledge graph and web-based application that integrates selected gene set libraries from Enrichr for comprehensive analysis and visualization. In Enrichr-KG, nodes represent genes or functional terms, while edges connect genes to their corresponding enriched terms. This network-based approach reveals previously hidden relationships between genes and annotated terms across various datasets. Enrichr-KG currently encompasses over 20 gene set libraries, including categories such as transcription, pathways, ontologies, diseases, drugs, and cell types. For our study, we specifically used data from the KEGG 2021 HUMAN [17], Reactome 2022 [18], and Gene Ontology Biological Processes 2021 (GO-BPs) (Gene [19]).

2.6 | In Silico Drug Repurposing

For the in silico drug repurposing, we used a computational tool called DGB (https://maayanlab.cloud/DGB/) (accessed in September 2024) [10]. DGB is a web-based platform that leverages a comprehensive database of drug-induced transcriptomic profiles to rank drugs and small molecules based on their ability to modulate the expression of user-specified target genes. Users input a gene symbol and specify whether they aim to upregulate or downregulate its expression. The tool then generates a ranked list of small molecules that have experimentally been shown to cause the desired gene expression changes. DGB provides metrics for each molecule, such as $\log_2$FC, p value, and q value, which measure the effect size and statistical significance of the expression changes. The data in DGB is derived from the LINCS L1000 dataset [20], the Connectivity Map (CMap) [21], and the Gene Expression Omnibus (GEO) [22]. In our analysis, we prioritized small molecules with a q value < 0.05 and $|\log_2\text{FC}| \geq 1$ that effectively reversed the expression of our target genes within the L1000 dataset.

3 | Results

3.1 | Significant Altered and Monotonically Expressed Genes for Flight and Postflight

The first part of this study involved identifying the significantly altered genes in both flight and postflight conditions. From the differential expression analysis, we selected genes with p values < 0.05 as significant, resulting in 1026 significantly altered genes in the flight condition and 2734 in the postflight condition (Figure 2). The complete lists of these significant genes for both conditions are provided in Supporting Information File S1.

In the second part of our study, we focused on significantly altered genes that displayed a consistent monotonic trend (either consistently increasing or decreasing $\log_2$FC relative to ground control) across both space conditions. We specifically examined the 474 significant genes shared between flight and postflight conditions. For each gene, we compared the absolute $\log_2$FC between conditions, retaining those with the same directional trend. As shown in Figure 2, 130 of these common genes exhibited a higher $|\log_2\text{FC}|$ in flight than in postflight, while 343 genes showed a higher $|\log_2\text{FC}|$ in postflight compared to flight (one gene was excluded as it was present in opposite differential expression state between the two conditions). The monotonically expressed genes across both flight and postflight conditions are listed in Supporting Information File S2.

3.2 | Enriched Biological Terms of Monotonically Expressed Genes in Flight and Postflight Condition

Multisource enrichment analysis was conducted using Enrichr-KG to identify statistically significant enriched biological terms (BTs) involving the monotonically expressed genes in postflight condition. Using a q value < 0.05 as the selection criterion, we identified 13 enriched BT in the postflight condition: Metallothioneins Bind Metals R-HSA-5661231, Response To Metal Ions R-HSA-5660526, cellular response to cadmium ion (GO:0071276), cellular zinc ion homeostasis (GO:0006882), response to cadmium ion (GO:0046686), response to zinc ion (GO:0010043), zinc ion homeostasis (GO:0055069), cellular response to zinc ion (GO:0071294), negative regulation of growth (GO:0045926), cellular response to copper ion (GO:0071280), response to copper ion (GO:0046688), cellular response to metal ion (GO:0071248), and cellular transition metal ion homeostasis (GO:0046916) (Figure 3).

3.3 | NRF2-Related Monotonically Expressed Genes in Postflight Condition

In the second part of the study, we compared the monotonically expressed genes in postflight condition with the NRF2 neighborhood. More specifically, we compared the 343 monotonically expressed genes that were found to have higher $|\log_2\text{FCs}|$ in postflight versus flight condition, with the 229 genes in the NRF2 network. We found that eight genes–four over-expressed and four under-expressed–were found to be common between monotonically expressed in the postflight condition and NRF2-related genes: *ZBTB24*, *BRD8*, *ATF7*, *TRIM66 (under-expressed)*, *GLRX*, *JUN*, *ATF3*, and *PRDX1 (over-expressed)* (Figure 4).

3.4 | Genetic Associations of the Monotonically Expressed Genes With CVD in Postflight Condition

Out of the information available from GWAS catalog, our analysis revealed associations of genome-wide significance (p value $< 5 \times 10–^{8}$) for CVD-related outcomes that involved six genes out

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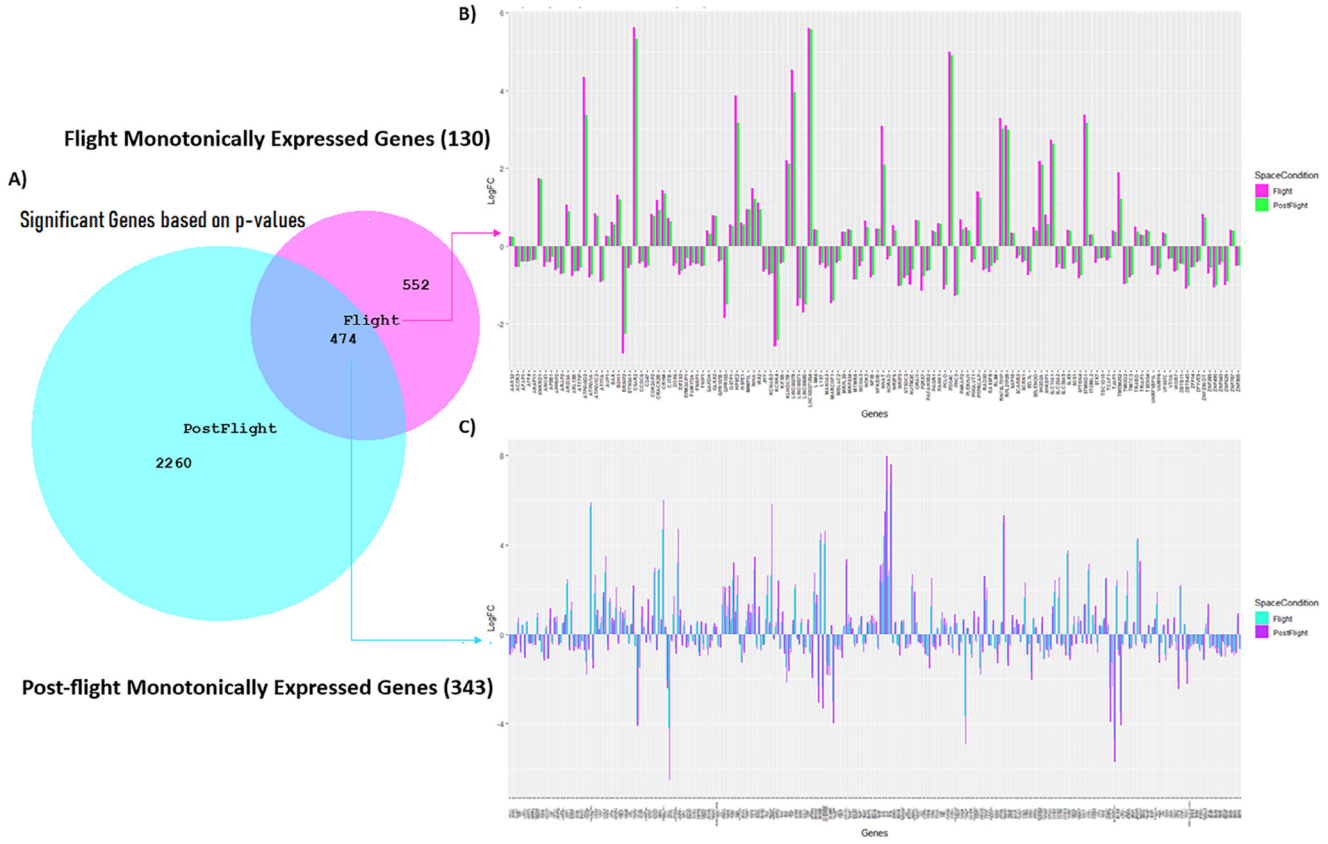


FIGURE 2 | (A) Venn diagram for the identification of the significant common genes between altered genes in flight and postflight. (B) Monotonically expressed genes with flight $|\log_2 FCs|$ greater than postflight $|\log_2 FCs|$. (C) Monotonically expressed genes with postflight $|\log_2 FCs|$ greater than flight $|\log_2 FCs|$.

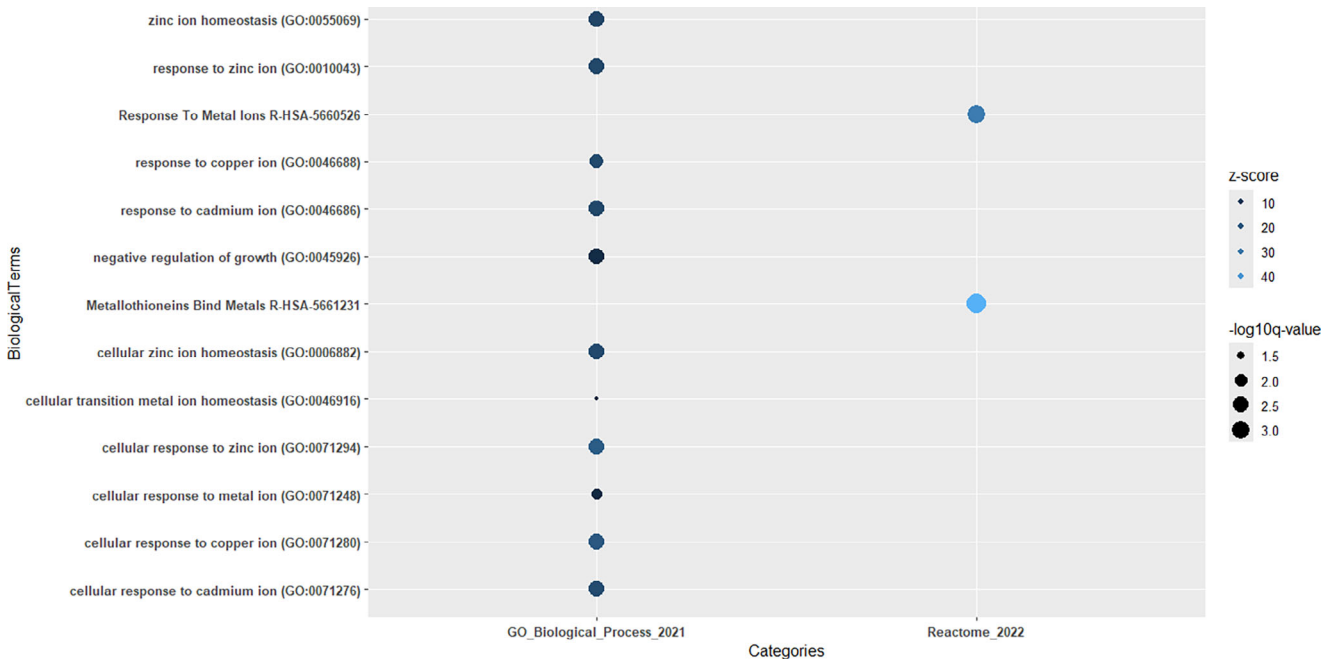


FIGURE 3 | Significantly enriched biological terms of the monotonically expressed genes in the postflight condition. The bubble size corresponds to –log10(q values) and the colors to the corresponding z-scores of the enrichment analysis procedure.

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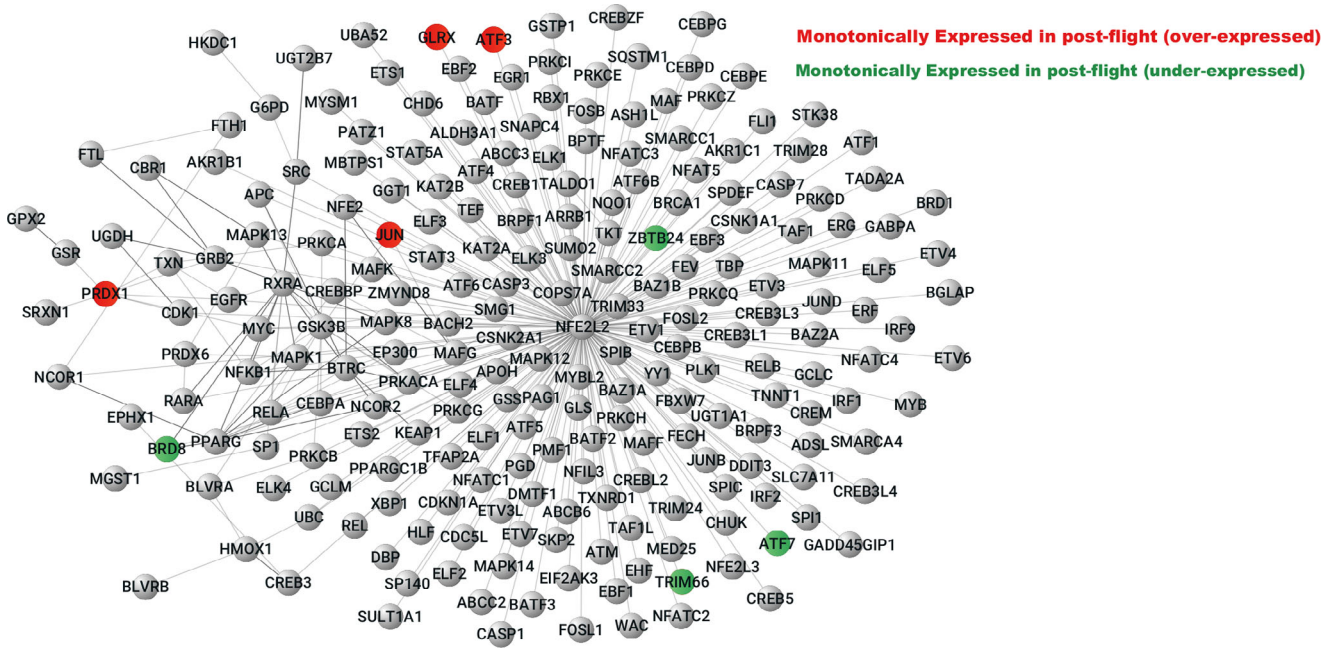


FIGURE 4 | Nuclear factor erythroid 2-related factor 2 (NRF2) interactome and regulome. Monotonically over-expressed genes in postflight are represented with red color monotonically under-expressed genes with green color.

of 343 that were used as input. The outcomes included QT interval, myocardial infarction, sudden cardiac arrest, peripheral artery disease, coronary artery disease, and cardiometabolic and hematological traits (Supporting Information File S3). Only one of these genes was linked to overexpression postflight, while the rest five were linked to under-expression. Four introns, two 5′ UTRs and one intergenic variant were linked to the CVD-related genes, while all of them but one were common (minor allele frequency [MAF] ≥ 5%). The only rare genetic variant identified (rs4665058, MAF = 0.014) was associated with sudden cardiac arrest (OR = 1.92, p value = 2×10^{-10}) and was linked to *BAZ2B* gene. We further identified associations for CVD-related risk factors including lipid biomarkers, such as total, LDL, and HDL cholesterol, as well as triglycerides, along with pulse pressure, systolic, and diastolic blood pressure (Supporting Information File S3). Eighteen genes were involved in the latter analysis with eleven of them linked to a risk >5% with CVD risk factors. Additionally, seven of them were overexpressed and eleven of them were under-expressed postflight. The genetic variants associated with the CVD risk factors included eighteen introns, one regulatory region, three intergenic, two noncoding transcript exon, one 3′ UTR, and one 5′ UTR variants. Similar to the CVD-related outcomes' analysis, only one genetic variant was rare (rs5763662, MAF = 0.04) that is linked to *MTMR3* gene and is associated with LDL cholesterol (OR = 1.08, p value = 1×10^{-8}).

3.5 | Identification of Candidate Repurposed Drugs for Monotonically Expressed Genes in Postflight

NRF2-related monotonically expressed genes as well as GWAS-related monotonically expressed genes in postflight were individually analyzed using the DGB drug repurposing tool. We applied filtering criteria of q value < 0.05 and an absolute $\log_2$FC ≥ 1 to identify potential drugs that regulate these genes. From the eight NRF2-related monotonically expressed genes, we found candidate repurposed drugs that reverse the expression of seven genes (*ZBTB24*, *BRD8*, *TRIM66*, *GLRX*, *JUN*, *ATF3*, and *PRDX1*). For the case of GWAs-significant monotonically expressed genes, we found candidate repurposed drugs for 19 out of 23 genes. Using a majority voting approach, we ranked the candidate drugs based on how many genes they influenced. From there, we selected the drugs that reversed the expression of at least 60% of the NRF2- and GWAS-related monotonically expressed genes in postflight condition, leading to the identification of 10 unique candidate repurposed drugs as presented in Figure 5.

3.6 | Computational Verification of the Shortlisted Repurposed Drugs

To further investigate the 10 candidate drugs, we examined their corresponding gene targets and modes of action (MoAs) using several databases, including DrugBank (https://go.drugbank.com/) [23] and CLUE–The Drug Repurposing Hub (https://clue.io/repurposing) [24] (Table 1). Additionally, we conducted an in-depth analysis of the pharmacological properties, chemical structures, and regulatory statuses of these drugs, leveraging data from the FDA database (https://www.fda.gov), DrugBank, PubChem (https://pubchem.ncbi.nlm.nih.gov/) [25], and SwissADME (http://www.swissadme.ch) [26]. To assess the drug candidates' potential for oral bioavailability, we evaluated their compliance with Lipinski's Rule of Five using SwissADME. A summary of these findings is presented in Table 1. Comprehensive results from the SwissADME analysis for all 10 repurposed drugs are provided in Supporting Information File S5.

We utilized the web tool Enrichr-KG to identify the pathways associated with the target genes of each drug candidate, focusing

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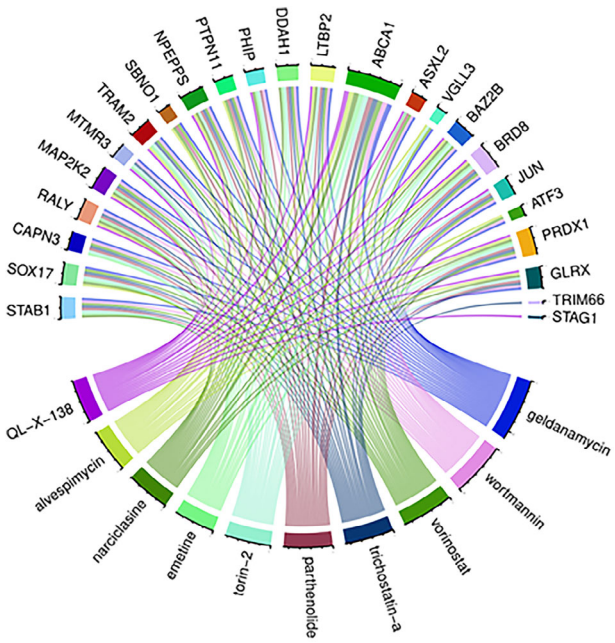


FIGURE 5 | Circos plot that summarizes the associations between the shortlisted repurposed drugs and the NRF2- and GWAS-related monotonically expressed genes. GWAS, genome-wide association studies; NRF2, nuclear factor erythroid 2-related factor 2.

on the KEGG 2021 HUMAN, Reactome 2022, and GO-BPs databases. Given the limited number of drug targets in many of the candidates, we did not apply a threshold for pathway selection and conducted a membership analysis. Ultimately, we included all pathways targeted by each drug in our final selection (Supporting Information File S4).

We searched in the MalaCards database (https://www.malacards.org/) [27] to find the molecular mechanisms and the biological processes that are associated with cardiovascular system disease. MalaCards is an integrated and searchable disease database that offers detailed, user-friendly information on all documented human diseases. It includes a wide range of gene-disease associations from curated and computational sources, along with data on disease-related mutations, phenotypes, pathways, drugs, and more. We compared the CVD-related BTs with those that are targeted from the 10 shortlisted repurposed drugs. We observed that four drugs (*torin-2*, *vorinostat*, *trichostatin-a*, and *parthenolide*) target eleven CVD-related BTs Alzheimer disease, Death Receptor Signaling R-HSA-73887, NF-kB Is Activated And Signals Survival R-HSA-209560, Notch signaling pathway, Notch-HLH Transcription Pathway R-HSA-350054, P75 NTR Receptor-Mediated Signaling R-HSA-193704, p75NTR Negatively Regulates Cell Cycle Via SC1 R-HSA-193670, p75NTR Signals Via NF-kB R-HSA-193639, Regulated Proteolysis Of p75NTR R-HSA-193692, regulation of gene expression (GO:0010468), and Signaling By NOTCH R-HSA-157118.

We also investigated the regulatory interactions between the NRF2-related monotonically expressed genes, the GWAS-related monotonically expressed genes, and the drug-gene targets of the shortlisted candidate repurposed drugs. These elements were incorporated into the SIGNOR 3.0 (The SIGnaling Network Open Resource) database (https://signor.uniroma2.it/) [28], a repository containing manually curated causal relationships between human proteins, biologically significant chemicals, stimuli, and phenotypes. After querying the SIGNOR 3.0 database, regulatory interactions were returned for 4 out of the 10 candidate drugs. This is due to the fact that the database contains curated interaction data for only a subset of compounds. This step enabled us to investigate the regulatory interactions among four candidate drugs, with the results reflected in Figure 6. As shown in Figure 6, *trichostatin-a* and *vorinostat*, two of our shortlisted repurposed drugs, inhibit *HDAC3*. This inhibition reduces the activity of *HDAC1*, which otherwise negatively influences the expression of the target gene *RELA*. *RELA* upregulates the activity of *JUN*, a NRF2-related gene that is monotonically expressed. *JUN* was found to be more overexpressed in postflight conditions compared to in-flight conditions. Furthermore, the target gene *MTOR* is directly inhibited by the candidate repurposed drug *torin-2*, and indirectly inhibited by *wortmannin* through the suppression of *PIK3CA*. Ultimately, *MTOR* expression is downregulated by MTMR3, a GWAS-associated gene with monotonic expression. *MTMR3* was found to be significantly under-expressed in postflight conditions compared to in-flight conditions. Notably, *MTMR3* downregulates the expression of *MTOR*.

4 | Discussion

Spaceflight presents unique challenges to human health due to exposure to increased levels of cosmic radiation, microgravity, and associated oxidative stress [29]. These environmental factors can lead to cellular damage, inflammation, and a range of health complications, including cardiovascular issues, immune system impairment, and an elevated risk of cancer [30]. NRF2 is a critical transcription factor that regulates the body's defense mechanisms against oxidative stress by promoting the expression of antioxidant enzymes [31]. In recent years, there has been growing interest in NRF2 activators, such as sulforaphane, curcumin, and resveratrol, for their potential to enhance the NRF2 pathway and improve antioxidant responses. These compounds may offer protective benefits for astronauts by mitigating oxidative damage and inflammation during long-duration space missions. Therefore, exploring NRF2 activators as potential treatments could contribute significantly to safeguarding astronaut health in the challenging environment of space [4].

In this study, we developed a computational methodology to investigate the plasticity of gene expression profiles from flight to postflight conditions. Using RNA-seq data from hiPSC-CMs, we identified monotonically expressed genes, defined as DEGs that exhibit a consistent ascending or descending trend in their $\log_2$FCs across flight and postflight conditions. Enriched BTs of monotonically expressed genes were also identified. We then correlated the monotonically expressed genes in the postflight condition with GWAS-based analyses to identify those associated with CVD-related outcomes. Additionally, the monotonically expressed genes in postflight were mapped within the NRF2 network, including both interactome and regulome, to assess their links to redox signaling pathways. Finally, through an in silico drug repurposing pipeline, we identified candidate drugs with experimental evidence showing a significant impact on

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TABLE 1 | Overview of gene targets, moas, and pharmacological properties of the 10 repurposed drugs.

Drugs	MoAs	Target genes	Smiles	FDA	Description	Lipinski rules of 5
Geldanamycin	HSP inhibitor	HSP90AA1	C[C@H]1C[C@@H]([C@@H]([C@H](/C=C(/[C@@H]([C@H](/C=C\C=C(\C(=O)NC2=CC(=O)C(=C(C1)C2=O)OC)/C)OC)OC(=O)N\C)C)O)OC	No	Experimental HSP90 inhibitor used in preclinical studies for cancer research.	No; 2 violations: MW > 500, N or O > 10
Wortmannin	PI3K inhibitor	PI4KA|PI4KB|PIK3CA|PIK3CD|PIK3CG|PIK3R1|PLK1|PRKDC	CC(=O)O[C@@H]1C[C@]2([C@@H](CCC2=O)C3=C1[C@]4([C@H](OC(=O)C5=COC(=C54)C3=O)COC)C)C	No	Experimental PI3K inhibitor, primarily used in research on signaling pathways.	Yes; 0 violation
Vorinostat	HDAC inhibitor	HDAC1|HDAC10|HDAC11|HDAC2|HDAC3|HDAC5|HDAC6|HDAC8|HDAC9	C1=CC=C(C=C1)NC(=O)CCCCCCC(=O)NO	Yes	FDA-approved for the treatment of cutaneous T-cell lymphoma (CTCL).	Yes; 0 violation
Trichostatin-a	HDAC inhibitor	HDAC1|HDAC10|HDAC2|HDAC3|HDAC4|HDAC5|HDAC6|HDAC7|HDAC8|HDAC9	C[C@H](/C=C(\C)/C=C/C(=O)NO)C(=O)C1=CC=C(C=C1)N(C)C	No	Experimental HDAC inhibitor used in epigenetics and cancer research.	Yes; 0 violation
Parthenolide	NFkB pathway inhibitor	REL|RELB|IKBKB|NFKB1|NFKB2|RELA	C/C/1=C\CC[C@@]2([C@H](O2)[C@@H]3[C@@H](CC1)C(=C)C(=O)O3)C	No	Natural NF-κB pathway inhibitor, studied for its antiinflammatory and anticancer properties.	Yes; 0 violation
Torin-2	mTOR inhibitor	MTOR	C1=CC(=CC(=C1)N2C(=O)C=CC3=CN=C4C=CC(=CC4=C32)C5=CN=C(C=C5)N)C(F)(F)F	No	Experimental mTOR inhibitor used in cancer and aging-related research.	Yes; 0 violation
Emetine	Protein synthesis inhibitor	RPS2	CC[C@H]1CN2CCC3=CC(=C(C=C3[C@@H]2C[C@@H]1C[C@@H]4C5=CC(=C(C=C5CCN4)OC)OC)OC)OC	No	Historically used as an antiprotozoal agent, but not FDA-approved for modern clinical use.	Yes; 0 violation
Narciclasine			C1OC2=C(O1)C(=C3C(=C2)C4=C[C@@H]([C@H]([C@H]([C@@H]4NC3=O)O)O)O)O	No	Experimental compound with anticancer and antiinflammatory properties, studied mainly in preclinical models.	Yes; 0 violation
Alvespimycin	HSP inhibitor	HSP90AA1	C[C@H]1C[C@@H]([C@@H]([C@H](/C=C(/[C@@H]([C@H](/C=C\C=C(\C(=O)NC2=CC(=O)C(=C(C1)C2=O)NCCN(C)C)/C)OC)OC(=O)N\C)C)O)OC	No	Experimental HSP90 inhibitor (a derivative of geldanamycin) used in preclinical cancer studies.	No; 2 violations: MW > 500, N or O > 10
QL-X-138			CC1=C(C=C(C=C1)N2C(=O)C=CC3=CN=C4C=CC(=CC4=C32)C5=CNN=C5)NC(=O)C=C	No	Experimental compound under investigation; no specific clinical indications yet defined.	Yes; 0 violation

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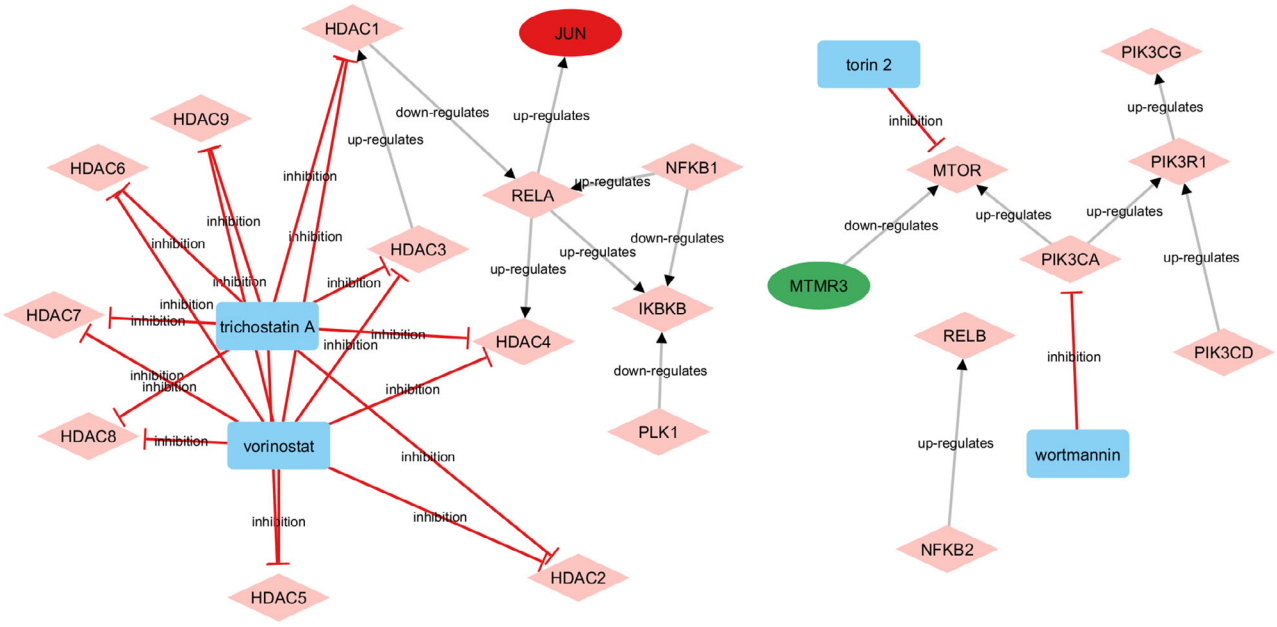


FIGURE 6 | The regulatory network comprises NRF2-related monotonically expressed genes, GWAS-related monotonically expressed genes, and drug-gene targets of the shortlisted candidate repurposed drugs. Gene targets are represented as rhombuses, NRF2-related monotonically over-expressed genes are shown in red ellipses, and GWAS-related monotonically under-expressed genes appear in green ellipses. GWAS, genome-wide association studies; NRF2, nuclear factor erythroid 2-related factor 2.

the expression of many GWAS- and NRF2-related monotonically expressed genes.

Summarizing the highlighted enriched BTs of the monotonically expressed genes in postflight, we found 13 molecular mechanisms and biological processes that are associated with metal ion. Metal ions play essential roles in numerous metabolic processes within the human body, making their homeostasis vital for sustaining life. In CVDs, the balance of metal ions is often disrupted, contributing to various physiological disturbances that can lead to impaired cardiac function.

Additionally, examining the NRF2-related monotonically expressed genes, we identified eight genes, four over-expressed and four under-expressed (*ZBTB24*, *BRD8*, *ATF7*, *TRIM66*, *GLRX*, *JUN*, *ATF3*, and *PRDX1*). It has been reported that *c-JUN* could regulate cardiomyocyte cell fate [32] and peroxiredoxins (Prdxs), regulate ROS levels balance against augmentation of ROS production during the pathogenesis of CVD [33]. Furthermore, *ATF3* expression in the heart is crucial for maladaptive responses. In cardiomyocytes, *ATF3* primarily regulates cardiac growth and triggers the fibrosis program in myofibroblasts, while in fibroblasts, *ATF3* promotes the hypertrophic gene expression in cardiomyocytes [34]. In the case of GWAS-related monotonically expressed genes, we found 22 CVD-related genes out of which 15 are associated with CVD outcomes and related risk factors with a risk ≥5% (*CCDC33*, *CAPN3*, *CSK*, *MAP2K2*, *MTMR3*, *SBNO1*, *PTPN11*, *PHIP*, *DDAH1*, *LTBP2*, *ABCA1*, *ANKRD9*, *ZNF37A*, *STAG1*, and *BAZ2B*). Using the 15 genes as input in GTEx (Genotype Tissue Expression Project v8) [35] within Expression Atlas [36], we found that *LTBP2* is highly expressed in aorta and coronary artery tissues (872 and 387 transcripts per million [TPM], respectively). *LTBP2* was under expressed postflight and is linked to systolic blood pressure (Supporting Information File S3). A mutation in *MAP2K2* has been reported as a cause of cardio-facio-cutaneous syndrome in an infant, leading to a severe and ultimately fatal progression of the disease [37]. Additionally, some MTMR family members have been reported to be differentially expressed in CVDs. Notably, *MTMR3* may serve as a potential therapeutic target in the treatment of myocardial fibrosis [38]. Furthermore, around half of Noonan syndrome cases are caused by missense gain-of-function mutations in the *PTPN11* gene, which encodes SHP2. This autosomal-dominant disorder, found in approximately 1 in 1000 to 1 in 2500 live births, often involves cardiac abnormalities such as pulmonary valve stenosis, septal defects, and hypertrophic cardiomyopathy [39]. Finally, *ABCA1* may help prevent CVD by reducing inflammation and supporting lipid balance. Studies suggest that posttranscriptional modifications are key in regulating *ABCA1* transport and localization to the plasma membrane, which is essential for its biological function [40].

The approach of the in silico drug repurposing resulted in 10 candidate repurposed drugs. Computational insights into the validity of these findings were performed as described in the Results section 3.6. Examining the results of the computational validity, three candidate repurposed drugs were found to be significant trichostatin-a, vorinostat, and torin-2. Trichostatin-a is a histone deacetylase inhibitor, that is, commonly utilized as an anticancer medication. It has been reported that trichostatin-a protects the heart from damage caused by oxidative stress [11]. In addition, in our previous work, trichostatin-a was found to be the top candidate repurposed drug that alters the expression of the most NRF2-related DEGs in AD. Through an experimental validation, we found that trichostatin-a activated a luciferase reporter for NRF2 activity and increased NRF2 protein levels in hippocampus-derived TH22 cells. This activation was further confirmed by the elevated expression of NRF2's downstream

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target, heme oxygenase 1, indicating an enhanced NRF2 transcriptional signature [41]. Moreover, vorinostat is also in the class of histone deacetylase inhibitors and it is used to treat cutaneous T-cell lymphoma. It has been reported that vorinostat helps prevent mitochondrial dysfunction and loss caused by ischemia-reperfusion injury and decreases myocardial ROS production when administered either before or after ischemia [42]. Additionally, torin-2 is a potent inhibitor of mTOR and exhibits anticancer properties. It is well known that mTOR signaling is crucial for maintaining cellular homeostasis; however, its abnormal activation is linked to numerous pathological conditions, including various cancers, metabolic, cardiovascular, and pulmonary diseases, as well as neurodegenerative disorders [43].

This study encountered several limitations that should be considered. First, the findings were derived from a small number of samples per condition, which does not fully capture the complexity of human physiology in space. Additionally, the short duration of exposure to spaceflight conditions does not adequately represent the long-term effects of spaceflight on cellular function. Therefore, extended studies are necessary to fully assess the impact of spaceflight on gene expression and physiological processes.

Moreover, the study's reliance on computational predictions introduces additional limitations. Although computational approaches are powerful tools for identifying candidate drugs and potential mechanisms of action, they are inherently constrained by the quality and completeness of the input data. In this case, the analysis depends on existing datasets and databases, which may not fully replicate the unique biological responses that occur in microgravity. For example, the accuracy of computationally predicted drug-target interactions and pharmacokinetic properties may vary, and these predictions do not account for a complete view regarding the complex interplay of multiple biological pathways under spaceflight conditions.

Although our study is limited by the absence of experimental evidence supporting our findings, some results have already been linked to spaceflight, suggesting the initial validity of our approach. Nevertheless, further investigation is required to explore additional associations. Overall, our findings highlight the potential of 10 shortlisted candidate drugs to address cardiovascular health challenges in spaceflight, with further validation required through in-vitro studies and preclinical animal models. The assessment of FDA approval status and compliance with Lipinski's Rule of Five underscores their feasibility for human use and oral bioavailability. However, further and extensive investigation is needed to evaluate their pharmacodynamic behavior in microgravity, safety for long-term missions, and potential drug–drug interactions.

To the best of our knowledge, no prior computational studies have specifically examined the monotonically expressed genes in flight and postflight condition in relation to the effects of spaceflight on human cardiac function. Despite the small sample size available, we anticipate that our findings will serve as prime candidates for future research and clinical experiments related to both flight and postflight conditions.

Author Contributions

Supervision of the study: George M. Spyrou. Conception and design of the study; collection, analysis, and interpretation of data; and drafting the article/revising it critically: All the authors.

Ethics Statement

The authors have nothing to report.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The original contributions presented in the study are included in the article/Supporting Information.

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